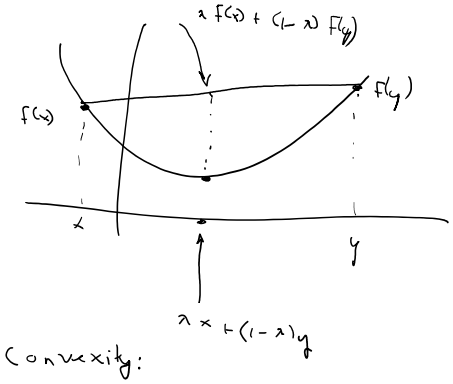



Convexity:

$$f(\lambda x + (1-\lambda) y) \le \lambda f(x) + (1-\lambda) f(y)$$

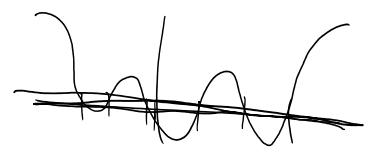
---

$$\underbrace{\|x\|_2 \le \|x\|_1}\ \le \sqrt{n}\,\|x\|_2$$

---

show: Lin indep: $1, t, t^2, \dots, t^m$

$$\underbrace{c_0 + c_1 t + c_2 t^2 + \dots + c_m t^m}_{p(t)} = 0$$



$$(1+x)^\alpha \qquad (1+x)^{m/n} \le 1 + \frac{m}{n} x$$

---

$$G(a_1, \dots, a_n) \le A(a_1, \dots, a_n)$$

Given $\left[\prod_{i=1}^{n} a_i\right]^{1/n} \le \frac{1}{n}\left[\sum_{i=1}^{n} a_i\right]$

Bounded: $x_n < M < \infty \ \forall n$

Unbounded: If there is a subsequence $x_{n_k}$ s.t.

$$\lim_{k\to\infty} x_{n_k} \to \infty$$

$$\big|\, \|x_n\| - \|x_m\| \,\big| \le \|x_n - x_m\|$$

---

$$\frac{1}{p} + \frac{1}{q} = 1$$

Given: $A^x B^y \le Ax + By$ $\qquad A(t) = |f(t)|^p \qquad B = |g(t)|^q$

$$\frac{1}{p} + \frac{1}{q} = 1$$

Given: $A^x B^y \le Ax + By$

$$A(t) = \frac{|f(t)|^p}{\int_a^b |f(t)|^p dt} \qquad B = \frac{|g(t)|^q}{\int_a^b |g(t)|^q dt}$$

obs: $\int_a^b A(t)\,dt = 1$ $\qquad \int_a^b B(t)\,dt = 1.$

$$\left[\frac{|f(t)|^p}{\int_a^b |f(t)|^p dt}\right]^x \cdot \left[\frac{|g(t)|^q}{\int_a^b |g(t)|^q dt}\right]^y \le \frac{|f(t)|^p}{\int_a^b |f(t)|^p dt}\, x + \frac{|g(t)|^q}{\int_a^b |g(t)|^q dt}\, y$$

Let $x = \frac{1}{p}$, $y = \frac{1}{q}$, $x + y = 1$

$$\int_a^b |f(t)| \cdot |g(t)|\, dt \le \int_a^b |f(t)|^p \left[\int_a^b |f(t)|^p dt\right]^{x-1} \left[\int_a^b |g(t)|^q dt\right]^{y \frac{1}{p}}$$

$$+ |g(t)|^q \left[\int_a^b |f(t)|^p dt\right]^x \left[\int_a^b |g(t)|^q dt\right]^{y-1 \frac{1}{q}} dt$$

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Goal: $$\left|\int_a^b f(t)\, g(t)\, dt\right| \le \left[\int_a^b |f(t)|^p dt\right]^{1/p} \left[\int_a^b |g(t)|^q dt\right]^{1/q}.$$

---

$$e^{jnt}, \quad n = 0, 1, 2, \ldots, N-1$$

$$c_0 e^{j0t} + c_1 e^{jt} + c_2 e^{j2t} + \cdots + c_{N-1} e^{j(N-1)t} = 0$$

$\frac{d}{dt}$ $\quad c_0 \cdot 0 + c_1 j e^{jt} + c_2 j2 e^{j2t} + \cdots + c_{N-1}(N-1)j\, e^{j(N-1)t} = 0$

$\frac{d^2}{dt^2}$ $\quad c_0 \cdot 0 + c_1 (j^2) e^{jt} + c_2 (2j)^2 e^{j2t} + \cdots + c_{N-1}[(N-1)j]^2 e^{j(N-1)t} = 0$

$\vdots$

$\frac{d^{N-1}}{dt^{N-1}}$ $\quad c_0 \cdot 0 + c_1 (j)^{N-1} e^{jt} + c_2 (2j)^{N-1} e^{j2t} + \cdots c_{N-1}[(N-1)j]^{N-1} e^{j(N-1)t} = 0$

$$\begin{bmatrix} e^{j0t} & e^{jt} & e^{j2t} & \cdots & e^{j(N-1)t} \\ 0e^{j0t} & je^{jt} & (2j)e^{j2t} & \cdots & (N-1)j\, e^{j(N-1)t} \\ & & \vdots & & \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{N-1} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

Let $d_0 = c_0 e^{j0t}$

$d_1 = c_1 e^{j1t}$

$d_2 = c_2 e^{j2t}$

$\vdots$

$d_{N-1} = c_{N-1} e^{j(N-1)t}$

$$\underbrace{\begin{bmatrix} \cdot & & & \\ & \cdot & & \\ & & \cdot & \\ & & & \cdot \end{bmatrix}}_{V} \begin{bmatrix} d_0 \\ d_1 \\ \vdots \\ d_{N-1} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$